

Session 8 :

Large Language Models (LLMs)

You've learned how AI can create stories, images, audio, etc. You've experimented with ChatGPT and Gemini to generate birthday artifacts. But have you ever wondered what's actually happening inside these chatbots? How do they know what to say? How can they sound so... human?

Today, we're opening up the box and discovering the fascinating technology behind AI conversation - Large Language Models.



Objectives

By the end of this session, you will be able to:

01

Explain what a language model is and how it predicts the next word using probabilities

02

Describe how LLMs are trained on massive amounts of text

03

Understand tokens - the building blocks of language for AI

04

Understand attention - how the model focuses on important words

05

Define parameters - the internal "knobs" that store patterns

06

Explain the context window - the model's short-term memory

07

Understand why LLMs sound so human (hint: it's patterns, not thinking)

08

List the main capabilities and limitations of LLMs

09

Compare how different LLMs (ChatGPT vs Gemini) respond to the same prompt

Recap : What We Learned in Session 7

Key Term	What It Means
Generative AI	AI that creates new content – text, images, audio – instead of just sorting or labeling
Generate	To produce something new based on patterns learned from data
Prompt	The instruction or question you give to an AI to tell it what to create

- ① Generative AI doesn't "understand" what it creates - it predicts what should come next based on patterns in its training data. You created a birthday artifact using ChatGPT or Gemini. Today, we'll dive deep into the technology behind the text you generated - **Large Language Models**.

The Big Question

"Have you ever wondered how ChatGPT or Gemini can have a conversation that feels so... human? Does it actually think? Does it have feelings? Or is it something else entirely?"

- ❏ **Today's Mystery:** We're going to open up the box and see what's really going on inside Large Language Models.





What is a Language Model?

A language model is an AI that **predicts the next word** in a sentence.

"The cat sat on the ____"

What word would you guess? Probably "mat" or "chair". How did you know? Because you've heard similar sentences before - you've learned the patterns of language. A language model does the **EXACT same thing** - but on a massive scale.

Your Brain Vs Language Model

Your Brain	Language Model
Learned from years of hearing language	Learned from billions of sentences
Predicts based on patterns AND meaning	Predicts based only on patterns
Understands what words mean	Doesn't understand - just matches patterns
Has feelings, memories, experiences	Has no feelings, no memories, no experience
Can write paragraphs because you understand the topic	Can write paragraphs by using patterns it has seen

❗ YOU know what you're writing about. The model doesn't. It has seen so many examples that it can imitate human writing perfectly - but there is no understanding inside.

Next-Word Prediction with Probabilities

A language model doesn't just guess one word - it **calculates probabilities** for every possible next word and picks the most likely one.

Example Prompt: "The capital of France is"

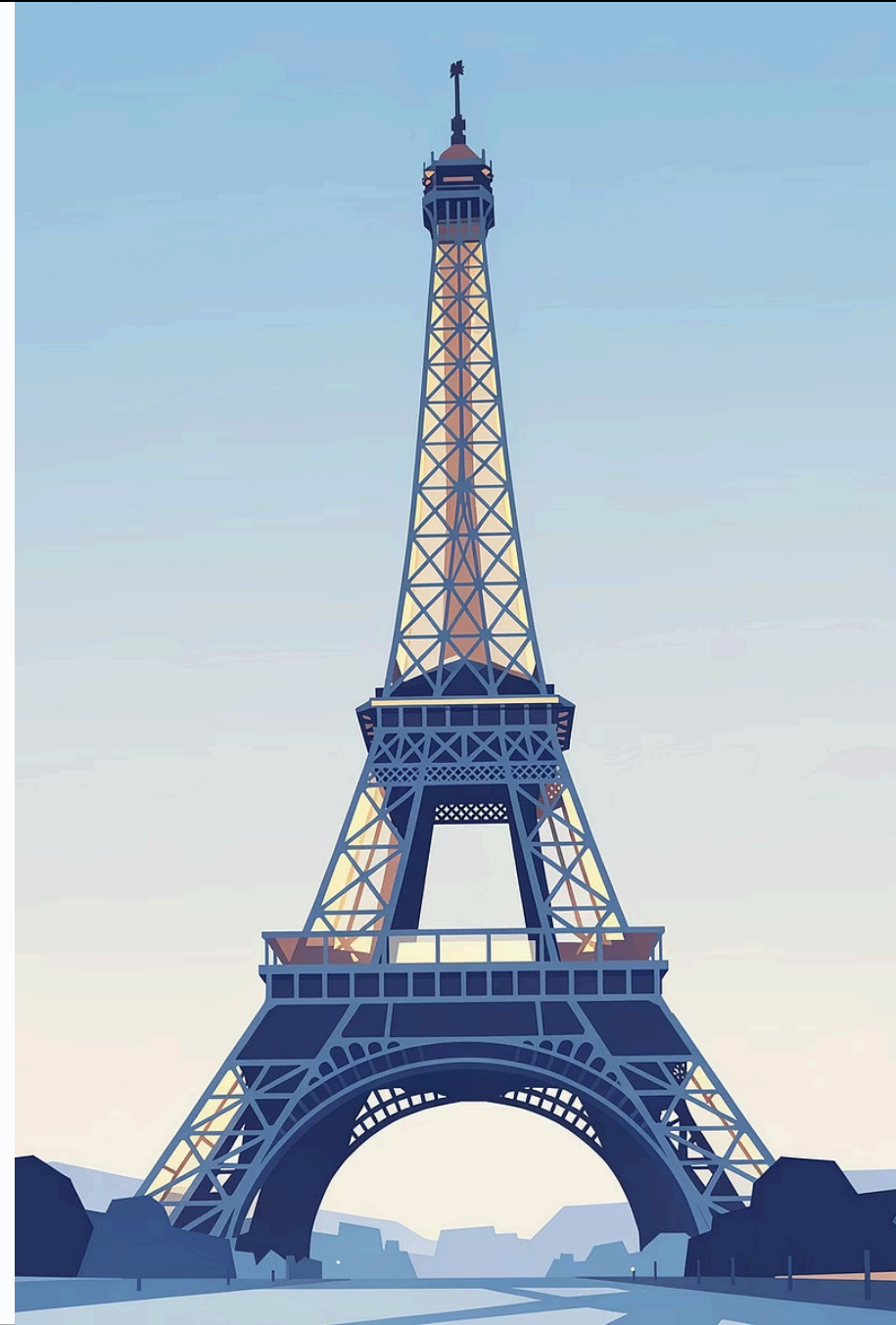
Next Word	Probability
Paris	97.3%
a	1.2%
the	0.8%
London	0.3%
city	0.2%
located	0.1%
(millions more...)	(tiny percentages)

The model chooses "**Paris**" because it has the highest probability - 97.3%. Then it repeats: *"The capital of France is Paris"* → Next prediction: What comes after "Paris"?

But if the prompt continued:

"Paris is famous for its _____" → New probabilities: "Eiffel Tower" (82%), "food" (7%), "art" (4%), "fashion" (3%)...

① **Key Point:** The model doesn't "know" that Paris is a city in France. It has just seen the pattern "capital of France is Paris" so many times that it's the most probable continuation.



Probability in Action - A Visual Walk-Through



This is the core loop of LLM response. Whether you ask for a poem, a code snippet, or an explanation of photosynthesis - the model is always doing the same thing: calculating probabilities and picking the next most likely word, one token at a time.

How Are LLMs Trained?

Imagine this : What if you had to read every book, every website, every Wikipedia article ever written? That's what we do to train an LLM.

1

Collect Data

Gather billions of sentences from books, websites, articles, and more

2

Hide a Word

Take a sentence and cover up one word (like a fill-in-the-blank)

3

Ask the Model to Guess

The model predicts the missing word with probabilities

4

Check the Answer

Compare the prediction to the real word

5

Adjust the "Knobs"

If wrong, the model tweaks its internal settings slightly to make the correct answer more probable next time

6

Repeat

Do this billions of times over weeks or months

Mini Activity 1 - Predict the Next Word

Instructions: For each sentence, write the word you think is most likely to come next. Then estimate its probability (what percentage of people do you think would guess that word?).

#	Sentence	Your Prediction	Your Probability Estimate
1	"Once upon a ____"		%
2	"I like to eat pizza with ____"		%
3	"To bake a cake, you need flour, eggs, and ____"		%
4	"The opposite of hot is ____"		%
5	"If you study hard, you will ____"		%
6	"She opened her umbrella because it was ____"		%
7	"In the morning, I brush my ____"		%

Answer Key (with approximate probabilities):

#	Most Likely Word	Probability	Other Possibilities
1	time	70%	day (15%), story (10%)
2	cheese	60%	pepperoni (20%), friends (10%)
3	sugar	50%	butter (25%), milk (15%)
4	cold	95%	freezing (3%), hot (1% – irony!)
5	succeed	50%	pass (30%), learn (15%)
6	raining	80%	rainy (15%), wet (3%)
7	teeth	85%	hair (10%), face (3%)

❶ **Discussion:** Your brain is also calculating probabilities - just unconsciously! This is exactly what LLMs do, but with billions of patterns.

Tokens - The Building Blocks

When we say "next word prediction," it's actually a tiny bit more complicated. LLMs don't always work with whole words - they work with **tokens**.

A **token** is a piece of a word.

Word	Tokens
moonlight	[moon, light]
jumped	[jump, ed]
friendship	[friend, ship]
can't	[can, 't]
pineapple	[pine, apple]
unbelievable	[un, believe, able]

Why tokens?

Handles new words

The model can understand words it has never seen by recognizing their pieces

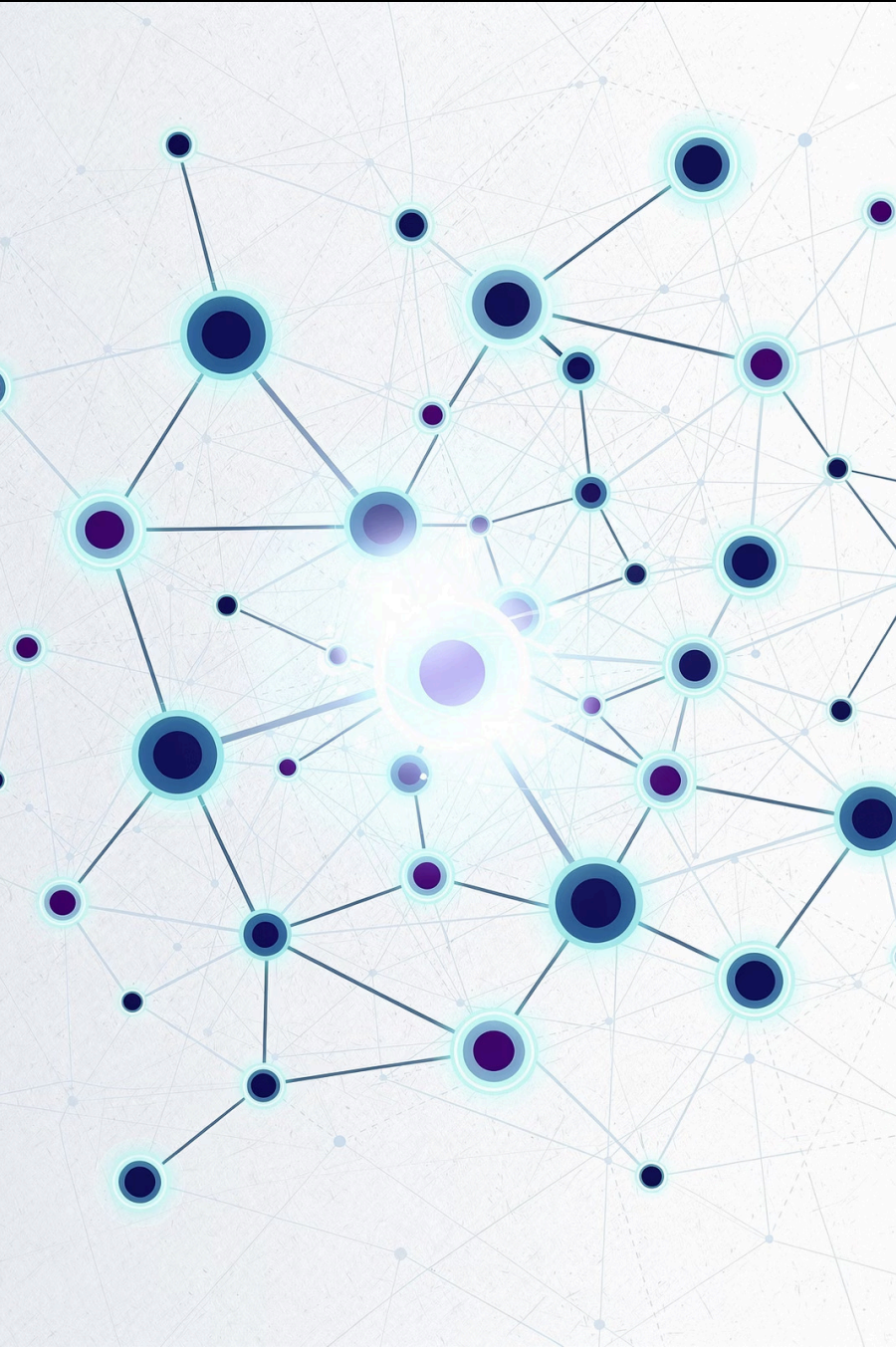
Efficient

Better to store 50,000 common pieces than millions of whole words

Works across languages

Languages like German that combine words work better with tokens

① Think of tokens like **Building Blocks**. With a set of blocks, you can build anything - even things you've never built before. Tokens work the same way for AI.

A network diagram with nodes and connections, featuring a central bright spot.

Attention (The Spotlight)

Imagine reading a long sentence. Not every word is equally important for understanding the meaning.

"The cat, which we adopted from the shelter last year and who loves to sleep in sunny spots, jumped onto the _____."

To predict the blank, which words matter most? "**cat**" - tells us the subject. "**jumped onto**" - tells us an action is happening. The long description in the middle? Less important for this prediction.

Attention is a mechanism that lets the model focus on the most important words - like a spotlight or a highlighter.

Without Attention

- All words treated equally
- Gets confused by long sentences
- Misses connections between distant words

With Attention

- Important words get "highlighted"
- Can pick out what matters
- Connects "cat" to "jumped" even with many words in between

① **Why It's a Breakthrough:** Before attention, AI models would "forget" the beginning of a long sentence by the time they reached the end. Attention allows them to look back at important words, no matter how far away.

Parameters

Parameters are the **internal numbers** inside an AI model that store everything it learns during training. Think of them as the model's brain cells or memory knobs.

Model Size	Parameters	Analogy
Small model	Millions	Insect brain - simple tasks
Medium model	Billions	Small mammal - more capable
Large model	Hundreds of billions	Human-like scale (but not human-like thinking!)

  **Fun Fact:** GPT-3 (the model behind the first ChatGPT) has **175 billion parameters**. That's more than the number of stars in the Milky Way galaxy!

Bigger Models

- Can learn more patterns
- More accurate on complex tasks
- Need more data and computing power
- Expensive to build

Smaller Models

- Train faster
- Use less electricity
- Can run on phones
- Cheaper to run

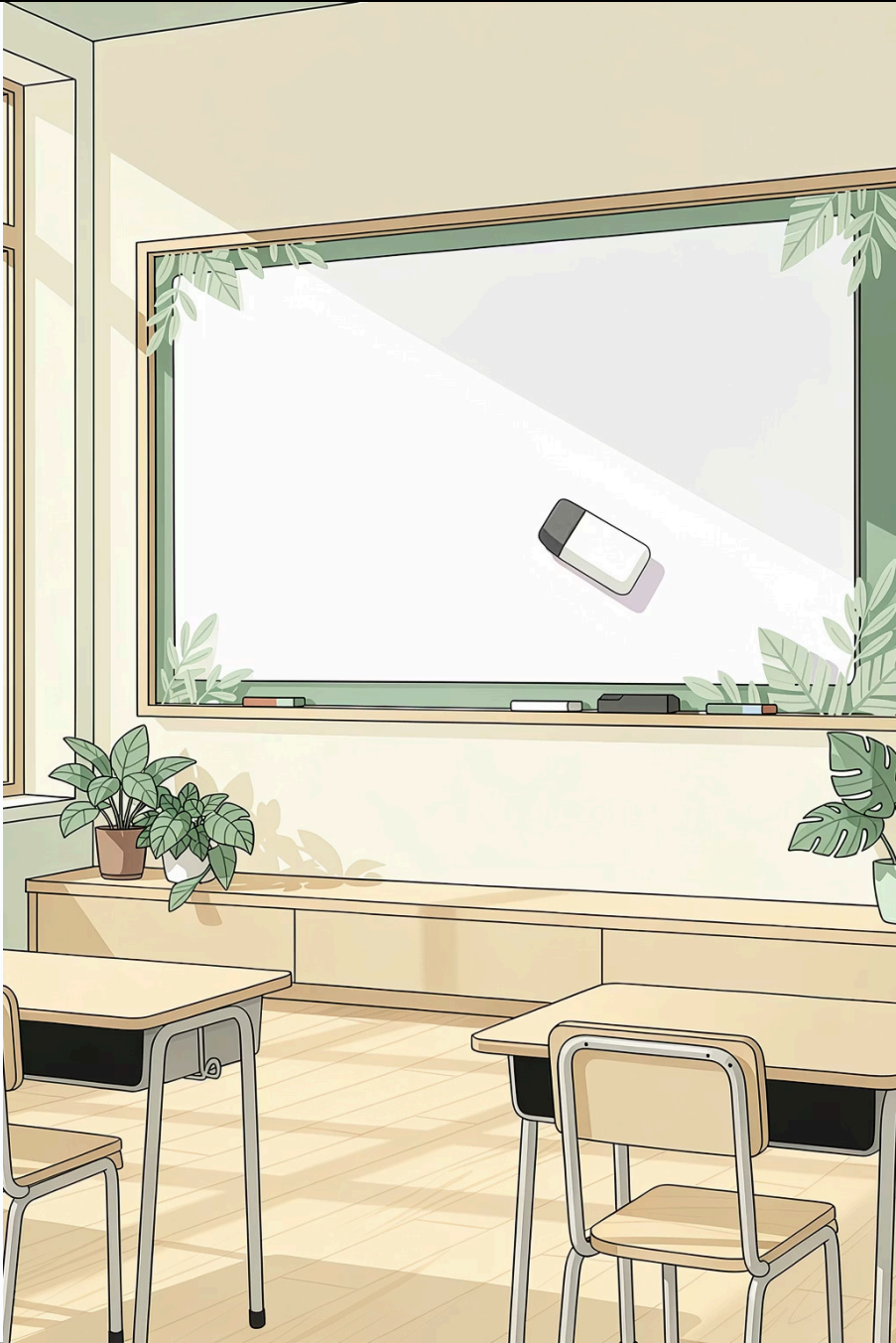
① **Key Insight:** A well-designed smaller model can sometimes outperform a sloppy large one.

Context Window

When you chat with an LLM, it doesn't remember everything forever. It has a **context window** - its short-term memory.

The **context window** is the maximum amount of text (in words or tokens) the model can "see" at once when generating a response.

Model	Typical Context Window
Older models	2,000–4,000 words
Newer models	8,000–100,000+ words
Gemini 1.5 Pro	Up to 1 million tokens!



Why this matters:

Situation	What Happens
Long conversation	Model forgets what you said at the beginning
Uploading a long document	Only part fits; ask about specific sections
Writing a long story	Model might forget character names from earlier chapters

❗ **Analogy:** It's like a whiteboard that can only hold a certain number of words. When the whiteboard fills up, you must erase old stuff to write new stuff - and anything erased is gone.

Mini Activity 2 - Token Counting Challenge

Instructions: Count how many tokens you think each sentence contains. Remember, tokens are pieces of words - not necessarily whole words!

#	Sentence	Your Token Count	Why?
1	"The rocket launched into space."		
2	"My dog loves to chase squirrels."		
3	"I'm learning to play guitar."		
4	"Wow, that was incredible!"		
5	"Don't forget your jacket."		
6	"She's the fastest runner on the team."		
7	"Let's meet at the library after school."		

Answer Key (approximate token counts):

#	Sentence	Tokens	Explanation
1	"The rocket launched into space."	6	Each word + the period = 6 pieces
2	"My dog loves to chase squirrels."	7	Each word + the period = 7 pieces
3	"I'm learning to play guitar."	7	"I" + "'m" = 2 tokens; then "learning", "to", "play", "guitar", "." = 5 more; total 7
4	"Wow, that was incredible!"	6	"Wow", ",", "that", "was", "incredible", "!" = 6 pieces
5	"Don't forget your jacket."	6	"Don" + "'t" = 2; then "forget", "your", "jacket", "." = 4 more; total 6
6	"She's the fastest runner on the team."	9	"She" + "'s" = 2; then "the", "fastest", "runner", "on", "the", "team", "." = 7 more; total 9
7	"Let's meet at the library after school."	9	"Let" + "'s" = 2; then "meet", "at", "the", "library", "after", "school", "." = 7 more; total 9

① **Note:** Punctuation marks like periods, commas, and exclamation points count as their own tokens. Token counts can vary slightly between different AI systems, but these are typical splits.

Why Do LLMs Sound So Human?

This is the question everyone wonders: *If it's just predicting words, why does it feel like I'm talking to a person?*

1. Human Training Data

LLMs are trained on text written by humans - books, articles, conversations. They learn our patterns, our phrases, our ways of expressing ideas.

2. Conversation Patterns

The training data includes millions of dialogues. The model learns how humans greet, respond, ask questions, and say goodbye.

3. Emotional Language

Humans use emotional words ("I feel," "that's amazing," "I'm sorry"). The model learns when these words typically appear.

4. Fluency

The model becomes incredibly fluent because it has seen so many examples of fluent writing.

❏ **But Here's the Truth:** The model does **NOT** feel emotions. It does **NOT** understand what it's saying. It does **NOT** have consciousness.

The Parrot Analogy

A parrot can mimic human speech perfectly. It might say "Hello" at exactly the right moment. But the parrot doesn't understand "Hello". LLMs are the same.

- ① LLMs are **incredibly convincing**, but there is **no mind, no feelings, no understanding** inside.



Capabilities of LLMs

What LLMs CAN Do:

Answer Questions

"What is the capital of Brazil?" → "Brasília"

Summarize

Paste a news article and ask for a one-paragraph summary

Translate

"Translate 'hello' to Spanish" → "hola"

Write Stories or Poems

"Write a haiku about a robot"

Explain Complex Topics

"Explain photosynthesis like I'm 10"

Generate Code

"Write Python code to sort a list"

Brainstorm Ideas

"Give me 10 creative business names for a pet store"

Role-play

"Act as a history teacher and quiz me on ancient Egypt"

❗ **The Amazing Thing:** All of these come from the same core skill - predicting the next word. The model has learned so many patterns that it can adapt to many task.

Limitations of LLMs

What LLMs CANNOT Do:

Limitation	What It Means	Example
No true understanding	The model doesn't "know" anything - it's just predicting words	Can explain gravity but doesn't "feel" it
Outdated knowledge	Trained on data up to a certain date	Won't know about events from last week
Can be biased	Repeats stereotypes from training data	May associate certain jobs with specific genders
Context window limits	Forgets old parts of conversation	Loses track of long discussions
Can generate harmful content	Might produce toxic text without safeguards	Could give dangerous advice
No common sense	Fails at obvious reasoning	Might say you can fry an egg on the sun
Can't verify facts	Doesn't check if information is true	Confidently states false "facts"

Hands-On Activity - Compare ChatGPT vs Gemini

Step-by-Step Instructions:

01

Open both tools

Have ChatGPT and Gemini ready in separate tabs

03

Ask the same prompts

Copy and paste exactly the same prompt to both

05

Record your observations

Use the table on the next page

02

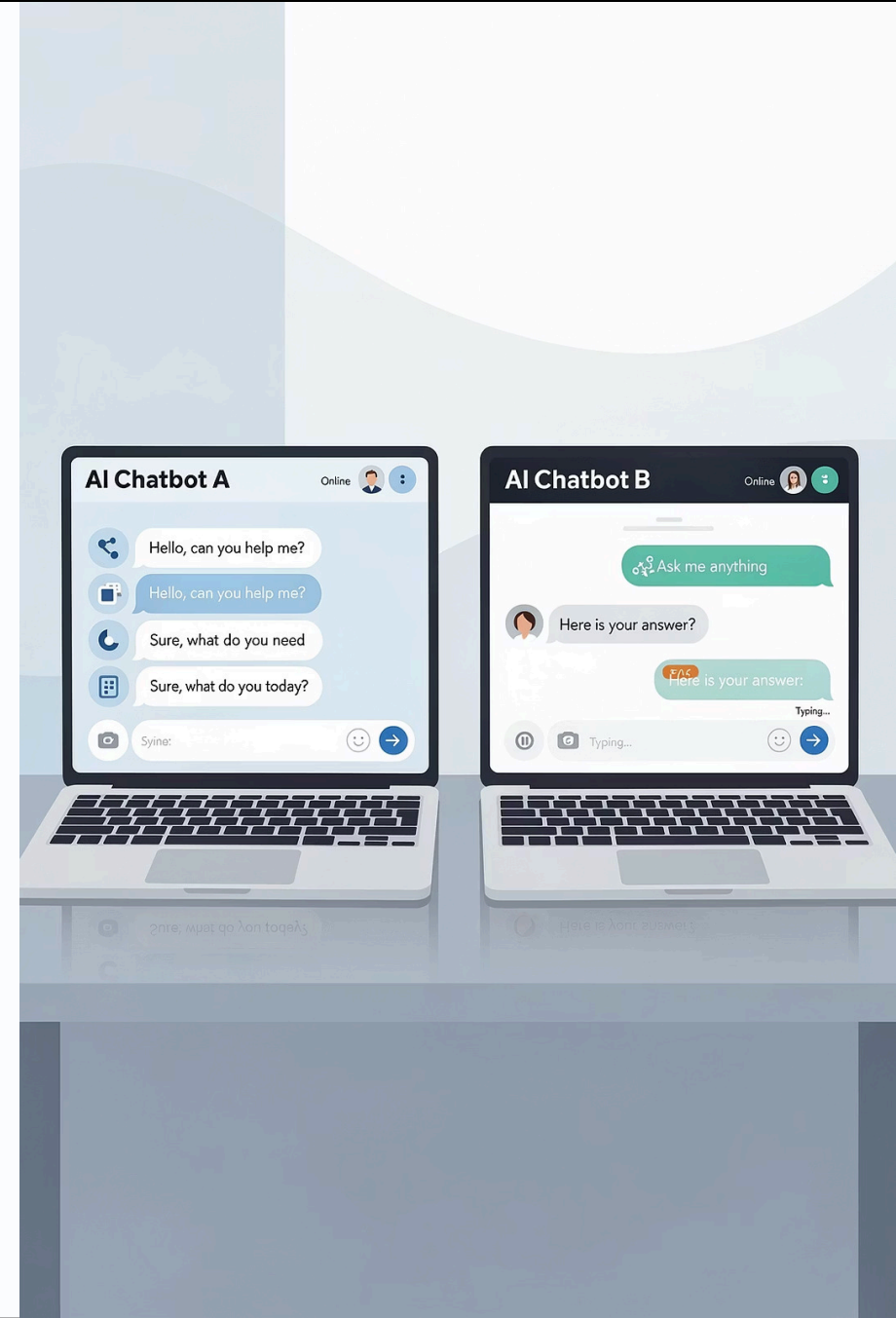
Start fresh

Begin new conversations in both to avoid memory from previous chats

04

Compare responses

Note the differences in tone, length, style, and content



Prompts to Try

1

"Tell me a fun fact about space."

2

"Write a short poem about a robot who loves to dance."

3

"Explain what AI is to a 5-year-old."

4

"Give me three ideas for a school science project."

5

"What's the best way to learn a new language?"

6

"Write a joke about computers."

7

"If you could travel anywhere in the world, where would you go and why?"

Questions to consider for each response:

- Which one is more creative?
- Which one is more detailed?
- Which one is more fun/engaging?
- Which one seems more accurate?
- Which one do you prefer for this prompt? Why?

Hands-On Activity - Comparison Table

Record your observations here:

Prompt #	ChatGPT's Response (Key features)	Gemini's Response (Key features)
1		
2		
3		
4		
5		
6		
7		

Hands-On Activity - Discussion

1 Did you notice any patterns?

Did one model consistently give longer responses?
More creative ones? More formal ones?

2 Why do you think they differ?

Both are trained on massive text data, but they may be trained on different datasets, have different "temperature" settings (creativity vs precision), or be optimized for different purposes.

3 Which model would you use for each task?

Writing a creative story? Getting help with homework? Having a casual chat? Explaining a complex topic?

4 Did either model say anything that seemed wrong or questionable?

If so, how would you check if it's correct?

 **Key Insight:** Different LLMs have different strengths. Choosing the right tool for the task is an important skill!

Summary

Concept	Summary
Language Model	Predicts the next word using probabilities
Training	Billions of fill-in-the-blank examples
Tokens	Pieces of words (LEGO blocks)
Attention	Spotlight on important words
Parameters	Brain cells that store patterns
Context Window	Short-term memory (whiteboard)
Why Human-like?	Learned from human text, but no understanding
Capabilities	Q&A, summarization, translation, writing, coding
Limitations	No understanding, outdated, biased, limited memory, hallucinations

 **Remember:** LLMs are probability calculators, not thinkers. They sound human because they learned from humans

Homework

Part 1: Vocabulary Matching

Match each term to its definition:

#	Term	Definition
1	Language Model	A. The internal "knobs" that store learned patterns
2	Parameters	B. How much text the model can remember at once
3	Context Window	C. An AI that predicts the next word
4	Token	D. A spotlight on important words
5	Attention	E. A piece of a word

Answers: 1-____, 2-____, 3-____, 4-____, 5-____

Part 2: Probability Practice

For each sentence, write the next word and estimate its probability:

Sentence	Next Word	Your Probability Estimate
"The movie was so boring that I fell ____"		%
"My phone battery is about to ____"		%
"To open the door, turn the ____"		%

Next Session Preview

Session 9 - AI Hallucinations

Have you ever seen an AI say something that was completely wrong but said it with total confidence? That's called a **hallucination**.

In Session 9, we'll explore:

